

#Q1. T-Test: Does gender affect teaching evaluation rates?

Why T-Test?

Independent variable: Gender → categorical, 2 groups (Male/Female) Dependent variable: Evaluation score (eval) → numerical Therefore, an independent two-sample t-test is appropriate. Hypotheses

H₀: There is no difference in teaching evaluation scores between male and female instructors.

H₁: There is a difference in teaching evaluation scores between male and female instructors.

```
import pandas as pd
df = pd.read_csv('/content/TeachingRatings.csv')
df
```

	rownames	minority	age	gender	credits	beauty	eval	division	native	tenure	students	all
0	1	yes	36	female	more	0.289916	4.3	upper	yes	yes	24	
1	2	no	59	male	more	-0.737732	4.5	upper	yes	yes	17	
2	3	no	51	male	more	-0.571984	3.7	upper	yes	yes	55	
3	4	no	40	female	more	-0.677963	4.3	upper	yes	yes	40	
4	5	no	31	female	more	1.509794	4.4	upper	yes	yes	42	
...	...	...	...	...	...	...	...	...	...	...	...	
458	459	no	32	male	more	1.231394	3.2	lower	yes	yes	9	
459	460	no	32	male	more	1.231394	4.3	upper	yes	yes	52	
460	461	yes	42	female	more	0.420400	3.3	upper	no	yes	52	
461	462	yes	42	female	more	0.420400	3.2	upper	no	yes	54	
462	463	yes	42	female	single	0.420400	4.1	lower	no	yes	28	

463 rows × 13 columns

Next steps: [Generate code with df](#) [New interactive sheet](#)

```
from scipy.stats import ttest_ind

female_eval = df[df['gender'] == 'female']['eval']
male_eval = df[df['gender'] == 'male']['eval']

t_stat, p_value = ttest_ind(
    female_eval,
    male_eval,
    equal_var=True
)

print("T-statistic:", t_stat)
print("P-value:", p_value)
```

T-statistic: -3.2499378389410034
P-value: 0.0012387613901376534

Decision

p-value = 0.00124

$\alpha = 0.05$

Since:

$0.00124 < 0.05$

we reject H_0 .

Conclusion

There is sufficient statistical evidence to conclude that gender affects teaching evaluation scores. The average teaching evaluation scores of male and female instructors are significantly different.

✓ Q2. ANOVA: Does beauty score differ by age?

age is a continuous numerical variable, while one-way ANOVA compares the means of 3 or more groups.

Therefore, we first divide age into three groups:

Age group

40 years and younger ≤ 40

Between 40 and 57 years >40 and <57

57 years and older ≥ 57

Hypotheses

H_0 : The mean beauty scores of all three age groups are equal.

$$H_0 : \mu_1 = \mu_2 = \mu_3$$

H_1 : At least one age group's mean beauty score is different.

```
#Create age groups
df.loc[df['age'] <= 40, 'age_group'] = '40 years and younger'

df.loc[
    (df['age'] > 40) & (df['age'] < 57),
    'age_group'
] = 'between 40 and 57 years'

df.loc[df['age'] >= 57, 'age_group'] = '57 years and older'
```

```
#Separate the Group
from scipy.stats import f_oneway

young = df[df['age_group'] == '40 years and younger']['beauty']

middle = df[
    df['age_group'] == 'between 40 and 57 years'
]['beauty']

old = df[
    df['age_group'] == '57 years and older'
]['beauty']
```

```
#perform Anova
f_stat, p_value = f_oneway(young, middle, old)
```

```
print("F-statistic:", f_stat)
print("P-value:", p_value)
```

```
F-statistic: 17.597558474277438
P-value: 4.322549530640899e-08
```

Decision

Since:

$$p < 0.05$$

we reject H_0 .

Conclusion

There is sufficient statistical evidence that beauty scores differ significantly among the different age groups. At least one age group's average beauty score is significantly different from the others.

✓ Q3. Chi-Square: Is there an association between tenure and gender?

Why Chi-Square?

Both variables are categorical:

gender → male/female

tenure → yes/no

Therefore, we use a Chi-Square Test of Independence.

Hypotheses

H_0 : Tenure and gender are independent.

H_1 : Tenure and gender are associated.

```
#Create contingency table
table = pd.crosstab(df['tenure'], df['gender'])

print(table)
```

```
gender  female  male
tenure
no         50    52
yes        145   216
```

```
#Perform Chi-Square test
from scipy.stats import chi2_contingency

chi2, p_value, dof, expected = chi2_contingency(table)

print("Chi-square:", chi2)
print("P-value:", p_value)
print("Degrees of freedom:", dof)
```

```
Chi-square: 2.20678166999886
P-value: 0.1374050603563787
Degrees of freedom: 1
```

Decision

$$0.1374 > 0.05$$

Therefore, we fail to reject H_0 .

Conclusion

There is no statistically significant association between tenure and gender in this dataset. In other words, the data do not provide sufficient evidence that tenure status is associated with gender.

✓ Q4. Correlation: Is teaching evaluation score correlated with beauty score?

Here both variables are numerical:

beauty $\rightarrow$ numerical eval $\rightarrow$ numerical

Therefore, we use Pearson correlation.

Hypotheses

H_0 : There is no correlation between beauty score and teaching evaluation score.

H_1 : There is a correlation between beauty score and teaching evaluation score.

```
from scipy.stats import pearsonr

correlation, p_value = pearsonr(
    df['beauty'],
    df['eval']
)

print("Correlation coefficient:", correlation)
print("P-value:", p_value)
```

```
Correlation coefficient: 0.18903909062278274
P-value: 4.247115507786e-05
```

Interpretation

The correlation coefficient is:

$$r = 0.189$$

This means there is a positive but weak correlation between beauty and teaching evaluation.

The p-value is:

$$p = 0.0000425 < 0.05$$

Therefore, we reject H_0 .

Conclusion

There is a statistically significant positive relationship between beauty score and teaching evaluation score. However, the correlation is weak, meaning beauty score has only a small linear association with teaching evaluation.

